

Table of Contents	Page
Introduction	1
Hardware	2
System Diagram	2
The Standard Module	3
Containers	4
Hubs	5
Firmware	6
Firmware Execution Flow	6
Module Firmware	7
CTRL	7
DSPL	7
Analog	7
I ² C Addresses	8
Assembly	9
Bill of Materials	9
Required Tools	13
Module Assembly	14
Faceplate Fabrication	14
Component Placement	15
PCB Soldering	17
Throttle Module	19
Telemetry Module	21
Joysticks	22
EVA Module	22
Adding the IC Chips	22
Container Assembly	23
Wiring	23
Schematics	24

Introduction

Untitled Space Craft is a brand of game controllers specialized for playing Kerbal Space Program and its derivatives, created by Codapop in 2021. This project is open source.

This document serves as a guide for those who choose to build their own USC controller.

The USC ecosystem is designed around a modular architecture. Individual control panels, referred to as *modules*, connect to a central *Hub* and communicate over a shared I²C bus. This allows builders to create anything from a small desktop controller with a single module to a full cockpit consisting of multiple containers and dozens of independent controls.

This manual covers the hardware, electronics, firmware, and assembly process required to build a USC controller. It includes mechanical drawings, schematics, bills of materials, firmware architecture, and assembly instructions. While basic soldering and electronics experience is recommended, every effort has been made to document the project thoroughly so that builders can understand not only how to assemble the controller, but also how it works and how to modify or extend it.